4 **(a)** **(i)** $I_1 = I_0 \cos^2\theta$

$$\frac{I_1}{I_0} = 0.50$$

$$\cos^2\theta = 0.50$$

$$\theta = \cos^{-1}\sqrt{0.50} = 45°$$

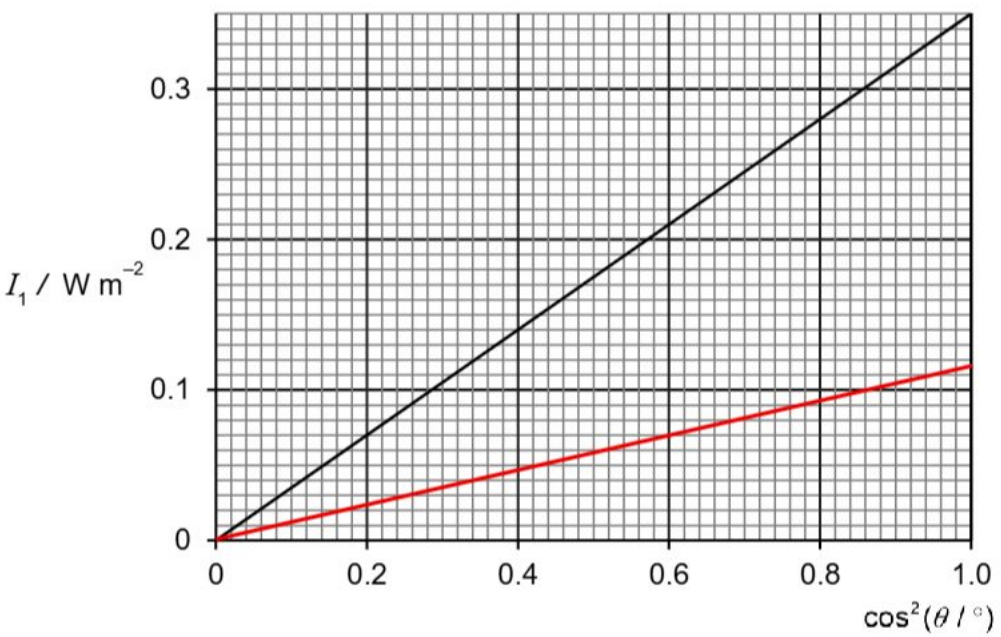
Or $180° - 45° = 135°$

Or $180° + 45° = 225°$

Or $360° - 45° = 315°$

Ans: 45°, 135°, 225° or 315° (any two)

(ii)



Gradient of I_1 with $\cos^2\theta$ gives the value of I_0
Draw straight line with gradient 1/3 of original line
(pass through origin and point (0.86, 0.100)

(b) $I \propto \frac{1}{r^2}$

Given $I_{20} = 20$ W m^{-2}

$$I_{40} = \left(\frac{r_{20}}{r_{40}}\right)^2 I_{20}$$

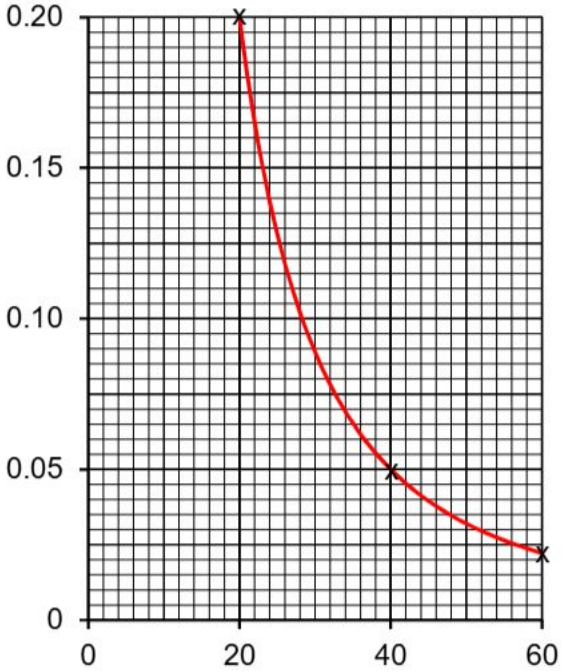
$$= \left(\frac{20}{40}\right)^2 (0.20)$$

$$= 0.050 \text{ W m}^{-2}$$

$$I_{60} = \left(\frac{r_{20}}{r_{60}}\right)^2 I_{20}$$

$$= \left(\frac{20}{60}\right)^2 (0.20)$$

$$= 0.022 \text{ W m}^{-2}$$



*smooth curve and all 3 points calculated and plotted correctly.